

Nathan Paul

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EXPERIENCE

Research Intern @ PES Innovation Lab (formerly Microsoft Innovation Lab) Jun '25 – Jul '25

- Worked on **MiCoz**, a research project exploring [tuning methodologies](#) for microservices
- Used virtual speedup to identify key bottleneck services across nodes.
- Prototyped latency instrumentation on real and simulated DAGs to pinpoint true bottlenecks.
- Presented research findings & optimization strategies to a professional audience at the final **MiM** event.

Platform Engineering Intern @ Golain.io. Remote, Pune Mar '25 – Apr '25

- Refactored the workflow/rule engine, migrating to a **DAG**-based approach for improved scalability
- Worked on rule logic and increased modularity across platform components

Developer, ACM Industrial Experience Program Feb '24 – Apr '24

- Part of the team that built **anna**, a static site generator written in Go.
- Achieved **284.5 ms** render time for a 1000-file [benchmark](#), outperforming Hugo and 11ty.
- Created a sitemap generator, set up RSS feeds, and an interactive tag-graph (D3.js).
- **DevOps**: Implemented CI/CD pipelines, a setup wizard, and theme bootstrapping.
- Presented at the [76th Go Bangalore meetup](#).

SKILLS & TECHNOLOGIES

Languages: Go, Python, Bash | **Linux, Docker, K8S, Git, Nix, SQLite, Redis, Postgres, eBPF**

PROJECTS

- [ash](#) — Event-driven Matrix bot framework with pluggable command system, AI integrations (Groq), and webhook support.
- [linkstash](#) — Link management and bookmarking service with tagging and categorization.
- [jukebox](#) — Headless music-queueing service that pulls LastFM recommendations and enqueues tracks into MPD using **yt-dlp**.
- [lava](#) — Node daemon that saves bookmarked articles into an Obsidian vault. Fetches, sanitises and organises content into markdown notes for offline reading and annotation.
- [wombat](#) — Wikipedia crawler that builds article graphs and visualises relationships between pages.
- [pasta](#) — Minimal pastebin service written in Go.
- [drago](#) — A proof-of-concept DAG-based workflow engine demonstrating lightweight scheduling and pluggable task runners.

FOSS CONTRIBUTIONS

- [anna](#): A lightning fast static site generator written in **Go**. Implements parallel rendering, live-reload and a minimal runtime; used to power fast personal sites and benchmarks where it outperformed Eleventy and Hugo.
- [elooop](#): A serverless all-in-one event-management system (RBAC, dynamic forms, QR checkins). We served 250+ concurrent users at our last hackathon. **NextJS, Google OAuth, Vercel, TursoDB, Tailwind**.

EDUCATION

B.Tech in Computer Science & Engineering at PES University 2023 – 2027

ISFCR Dean's Award: Ranked 6th out of 250 participants in a jeopardy-style CTF.

DAC (Distinction Scholarship): Merit-based scholarship for academic excellence in Sem 5.

Mentor Lead @HSP PESU June '25 – Present

- Leading the mentor-team and coordinating club-wide mentorship initiatives
- In charge of **Tilde 4.0**, HSP's flagship summer-mentorship program

Technical Mentor @HSP PESU Oct '23 – May '25

- Maintainer of the club [website](#), powered by Anna
- Mentored contributors to [psh](#), a POSIX-like shell
- Presented [jukebox](#) and **anna** at HSP Mentor Expo and Fireside Talks

Member @ACM PESU Sept '23 – Present

- Maintainer Lead for Hacknight 6.0
- Mentored contributors to [nimbus](#), a TUI client for Bluesky